

BUILD A CURSOR-TRACKING 3D PORTFOLIO HERO

A simple 3-step workflow: create the character → generate the animation → build the interactive hero.

STEP 01

Create the character
image

ChatGPT

STEP 02

Turn it into a
directional video

Google Flow

STEP 03

Build the complete
hero

Antigravity

sl_tech_journal • Full-stack developer • Freelancer

THE 3-STEP WORKFLOW

This guide turns the cursor-tracking hero from the reel into a repeatable workflow. Follow the steps in order and keep the prompts exactly as provided below.

01	IMAGE CREATION	Use ChatGPT. Upload your own photo first, then upload the provided character/reference image.
02	VIDEO CREATION	Use Google Flow. Give it the generated character image and paste the Step 2 animation prompt.
03	HERO BUILD	Use Antigramity. Put the generated image and video inside your project's public folder, open the

TOOLS USED

Step 1 — ChatGPT • Step 2 — Google Flow • Step 3 — Antigramity

Tip: If the first Antigramity attempt does not create the smooth cursor animation correctly, simply tell the AI to **fix the cursor animation** and refine the tracking until it is smooth.

STEP 01 — IMAGE CREATION

Tool: ChatGPT

Upload **your photo first**, then upload the provided image below as the second/reference image. Paste the prompt exactly.



REFERENCE IMAGE PROVIDED FOR STEP 01

Expected result: a polished 3D character image that keeps your identity recognizable while following the reference's style, pose, framing, lighting and red background.

STEP 01 — PROMPT

PROMPT — PART 1

Use this prompt with your photo as the first image and the uploaded image as the second/reference image:

Create a stylized 3D digital portrait using my uploaded photo as the PRIMARY IDENTITY REFERENCE and the second uploaded image as the STYLE, COMPOSITION, AND COLOR REFERENCE.

Preserve my exact facial identity, facial structure, skin tone, hairstyle, and recognizable features from my photo. Do not replace or redesign my face.

Recreate the reference image's overall look: high-quality polished 3D/animated character portrait, expressive large eyes, soft realistic skin shading, detailed hair strands, subtle blush, natural glossy lips, elegant facial proportions, and a friendly confident expression.

Match the same framing and composition as the reference: centered head-and-shoulders portrait, looking slightly upward/to the side, with the subject occupying the central area and clean negative space around them.

Use a bright vivid red background with a smooth clean gradient and strong visual contrast against the subject. Keep the background minimal with no objects or text.

Make the result premium, polished, cinematic, highly detailed, smooth 3D-rendered illustration, while keeping my identity recognizable.

Do not copy the reference person's identity. Use only its visual style, pose, framing, lighting, colors, and artistic treatment.

STEP 02 — VIDEO CREATION

Tool: **Google Flow**

Use the character image generated in Step 1 as the visual reference. Paste the complete Step 2 prompt below to generate the directional character animation.

Goal: the character should smoothly follow a cursor around the screen, with continuous head + eye movement and a fixed camera/background.

STEP 02 PROMPT — PART 1

Use the provided character image as the exact visual reference.

Create a smooth interactive-character animation designed specifically for a website cursor-tracking hero section.

The character is a cute young woman with curly dark hair, fair warm skin, large expressive brown eyes, silver hoop earrings, and an off-white sweater. Show only her head and shoulders against a perfectly solid bright red background.

The camera must remain completely fixed and frontal.

The animation must demonstrate CONTINUOUS directional head and eye movement, as if the character is naturally following a cursor moving freely around her.

Create smooth transitions through all directions:

center → up → upper-right → right → lower-right → down → lower-left → left → upper-left → center.

Also include smooth diagonal positions between every major direction.

The head movement must NOT jump between poses. Every directional change must blend smoothly and naturally, with subtle eye movement leading the head movement.

The character should be able to look:

- directly forward
- slightly up
- slightly down
- slightly left
- slightly right
- upper-left
- upper-right
- lower-left
- lower-right

STEP 02 PROMPT — PART 2

Make the transitions fluid and continuous so that intermediate cursor positions would naturally correspond to intermediate head positions.

Keep the character's face, hair, eyes, clothing, proportions, lighting and identity perfectly consistent throughout.

Only the eyes, head and very subtle hair movement should animate.

No talking.

No lip movement.

No body movement.

No hand movement.

No camera movement.

No zoom.

No background movement.

The bright red background must remain perfectly flat and uniform.

The animation should feel like a premium interactive portfolio website character that is continuously looking toward the user's mouse cursor.

Smooth cinematic 3D cartoon animation, high-quality facial consistency, natural head rotation, subtle eye tracking, soft realistic motion, seamless directional transitions.

STEP 03 — BUILD THE COMPLETE HERO

Tool: **Antigramity** (recommended for this workflow)

Before pasting the master prompt, prepare your project exactly like this:

1	Create/open your project root folder.
2	Inside it, create a folder named <code>public</code> .
3	Place the generated character image inside <code>public</code> .
4	Place the generated video inside <code>public</code> as <code>character.mp4</code> .
5	Open the root folder with your AI coding agent.
6	Paste the complete Step 3 master prompt.

If the AI creates the page but the cursor tracking is not smooth on the first attempt, tell it: **“Fix the cursor animation.”** Then ask it to correct the tracking/transition behavior rather than rebuilding the whole concept.

STEP 03 — MASTER PROMPT

MASTER PROMPT — PART 1

master prompt

I have placed a character animation video in `public/character.mp4`. The video features a character against a solid background whose head turns in 8 directional poses (clockwise/counter-clockwise), with a neutral pose looking directly at the camera at the end.

I want you to build an award-winning, luxury portfolio hero section with an ultra-smooth, zero-lag, zero-ghosting cursor-tracking character animation.

CRITICAL TECHNICAL CONSTRAINTS (DO NOT VIOLATE):

1. ****DO NOT USE CSS 3D TRANSFORMS ON THE PAGE/VIDEO:**** Never use `perspective`, `rotateX`, or `rotateY` on the canvas or container. The page and character's body must stay 100% rock-solid motionless. ONLY the character's head should rotate.
2. ****DO NOT SEEK OR PLAY THE MP4 VIDEO AT RUNTIME IN THE BROWSER:**** Generated MP4s have only 1 keyframe. In-browser `video.currentTime` seeking causes severe lag and freezes, and calling `video.play()` causes uncontrolled background playback.
3. ****PRE-EXTRACT WEBP FRAMES WITH PYTHON / OPENCV:****
 - Write a python script using `cv2` to inspect the video's frame count and timeline.
 - Identify the exact frame numbers for the 8 compass directions (UP, UP-RIGHT, RIGHT, DOWN-RIGHT, DOWN, DOWN-LEFT, LEFT, UP-LEFT) and the CENTER neutral frame.
 - Pre-extract 64 high-quality WebP frames along the 360° circular head rotation trajectory into `public/frames/` (each frame ~5.6° apart), plus `center.webp`.
4. ****ZERO-GHOSTING 60 FPS CANVAS RENDERER:****
 - Preload the frames in React into an image array.
 - In a `requestAnimationFrame` loop, calculate the cursor's angle relative to the character's face center (`atan2(dy, dx)`).
 - Use shortest-path circular angular lerp (`lerpAngle`) with a fast response factor (`~0.26`) so it tracks in ~35ms with zero lag.
 - Map the smoothed angle to the nearest frame index (0..63).
 - Draw ****EXACTLY ONE crisp frame at 100% opacity**** on the canvas. DO NOT alpha-blend overlapping frames (alpha blending creates double-face ghosting).
5. ****CENTER EYE CONTACT (DEADZONE):****
 - When the cursor is near the character's face (within ~12% screen radius), switch to `center.webp` so she looks directly into the user's eyes.
6. ****SEAMLESS BACKGROUND:****
 - Detect the exact RGB hex color of the video background and match the page and canvas background to it seamlessly.

MASTER PROMPT — PART 2

UI & HERO SECTION DESIGN:

- **Full Screen:** The canvas must cover the full viewport (`100vw`, `100vh`, `object-fit: cover`).
- **Header:** A floating, frosted-glass navigation pill centered at the very top (`[WORK]`, `[ABOUT]`, `[CONTACT]`) with `backdrop-filter: blur(20px)`.
- **Hero Typography (Bottom-Left):**
 - "Hi, I'm" in clean, spaced modern sans-serif.
 - Name (e.g., "Lohitha") in large, elegant cursive script (like Google Fonts 'Dancing Script') with a subtle soft drop shadow.
 - A compact 3-line bio about Full Stack Development (max-width ~340px) that does not crowd the character.
 - Two stylish white pill buttons: `Resume` (solid white with arrow icon) and `Let's Talk` (frosted glass/white border).
- **Custom Magnetic Cursor:** A glowing white custom cursor dot with a smooth trailing aura ring that scales up on interactive hover.

READY TO BUILD YOUR OWN?

I'm **sl_tech_journal** — a freelancer creating customized portfolios and website designs.

If you want me to handle your entire website — from design and animations to development — **DM me**.

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Workflow recap: **ChatGPT** → **Google Flow** → **Antigramity**

Save this guide and reuse the prompts whenever you want to build a cursor-reactive character hero.